

TD1:

```
To github.com:mrlnfroissart/esilv-marketpulse-g07-TortueNinja.git
* (new branch) marlon -> marlon
branch 'marlon' set up to track origin/marlon'.
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % pw
zsh: command not found: pw
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % pwd
/Users/marlon/labs/esilv-marketpulse-g07-TortueNinja
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % ls
CONTRIBUTING.md      TEAM_TEMPLATE.md      data                  labs                  requirements.txt
README.md             config                 evidence              readiness             src
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % ls -la
total 80
drwxr-xr-x  15 marlon  staff   480 Oct  7 18:28 .
drwxr-xr-x   4 marlon  staff   128 Oct  7 18:28 ..
-rw-r--r--   1 marlon  staff   8196 Oct  7 18:28 .DS_Store
drwxr-xr-x  12 marlon  staff   384 Oct  7 18:47 .git
-rw-r--r--   1 marlon  staff   178 Oct  7 17:56 .gitignore
-rw-r--r--   1 marlon  staff   2943 Oct  7 17:56 CONTRIBUTING.md
-rw-r--r--   1 marlon  staff   18278 Oct  7 17:56 README.md
-rw-r--r--   1 marlon  staff   1446 Oct  7 17:56 TEAM_TEMPLATE.md
drwxr-xr-x   3 marlon  staff    96 Oct  7 17:56 config
drwxr-xr-x   3 marlon  staff    96 Oct  7 17:56 data
drwxr-xr-x   3 marlon  staff    96 Oct  7 17:56 evidence
drwxr-xr-x   6 marlon  staff   192 Oct  7 17:56 labs
drwxr-xr-x   3 marlon  staff    96 Oct  7 17:56 readiness
-rw-r--r--   1 marlon  staff   133 Oct  7 17:56 requirements.txt
drwxr-xr-x   3 marlon  staff    96 Oct  7 17:56 src
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % cd data
(base) marlon@Marlons-MacBook-Air data % pwd
/Users/marlon/labs/esilv-marketpulse-g07-TortueNinja/data
(base) marlon@Marlons-MacBook-Air data % pwd
/Users/marlon/labs/esilv-marketpulse-g07-TortueNinja/data
(base) marlon@Marlons-MacBook-Air data % ls
sample
(base) marlon@Marlons-MacBook-Air data % cd sample
pwd
ls
/Users/marlon/labs/esilv-marketpulse-g07-TortueNinja/data/sample
bloomberg_reference.expected.json  bloomberg_reference_sample.json  instruments.json  prices.csv
(base) marlon@Marlons-MacBook-Air sample % cd ../..
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % pwd
/Users/marlon/labs/esilv-marketpulse-g07-TortueNinja
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % █
```

```
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38800000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.00,252.00,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.00,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44800000
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44800000
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % grep AAPL data/sample/prices.csv

2026-09-01,AAPL,249.20,251.50,248.00,250.00,38800000
2026-09-02,AAPL,250.00,252.00,249.60,251.30,39500000
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-04,AAPL,251.30,253.00,250.10,252.10,42500000
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44800000
2026-09-09,AAPL,254.00,255.70,252.80,254.70,38800000
2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000
2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000
2026-09-14,AAPL,256.60,258.30,255.40,256.00,42500000
2026-09-15,AAPL,257.20,259.50,256.00,258.00,44800000
2026-09-16,AAPL,256.70,258.70,255.50,257.20,38800000
2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000
2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000
2026-09-22,AAPL,261.10,262.80,259.90,261.30,44800000
2026-09-23,AAPL,261.20,263.50,260.00,262.00,38800000
2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000
2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000
2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44800000
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38800000
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % grep SP500 data/sample/prices.csv

2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,SP500,6623.00,6643.00,6611.00,6628.00,0
2026-09-09,SP500,6640.00,6657.00,6628.00,6642.00,0
2026-09-10,SP500,6642.00,6665.00,6630.00,6650.00,0
2026-09-11,SP500,6642.00,6662.00,6630.00,6647.00,0
2026-09-14,SP500,6663.00,6680.00,6651.00,6665.00,0
2026-09-15,SP500,6664.00,6687.00,6652.00,6672.00,0
2026-09-16,SP500,6663.00,6683.00,6651.00,6668.00,0
2026-09-17,SP500,6683.00,6700.00,6671.00,6685.00,0
2026-09-18,SP500,6684.00,6707.00,6672.00,6692.00,0
2026-09-21,SP500,6703.00,6720.00,6691.00,6708.00,0
2026-09-22,SP500,6703.00,6720.00,6691.00,6705.00,0
2026-09-23,SP500,6704.00,6727.00,6692.00,6712.00,0
2026-09-24,SP500,6715.00,6735.00,6703.00,6720.00,0
2026-09-25,SP500,6713.00,6730.00,6701.00,6715.00,0
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % █
```

```
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % python --version
Python 3.13.5
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % git --version
git version 2.54.0 (Apple Git-157)
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % python src/main.py

=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % █
```

```
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % grep -c AAPL data/sample/prices.csv
21
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % grep -c SP500 data/sample/prices.csv
21
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % ls -lh
total 48
-rw-r--r--  1 marlon  staff   2.9K Oct  7 17:56 CONTRIBUTING.md
-rw-r--r--  1 marlon  staff   10K Oct  7 17:56 README.md
-rw-r--r--  1 marlon  staff   1.4K Oct  7 17:56 TEAM_TEMPLATE.md
drwxr-xr-x  3 marlon  staff   96B Oct  7 17:56 config
drwxr-xr-x  3 marlon  staff   96B Oct  7 17:56 data
drwxr-xr-x  3 marlon  staff   96B Oct  7 17:56 evidence
drwxr-xr-x  6 marlon  staff  192B Oct  7 17:56 labs
drwxr-xr-x  3 marlon  staff   96B Oct  7 17:56 readiness
-rw-r--r--  1 marlon  staff  133B Oct  7 17:56 requirements.txt
drwxr-xr-x  3 marlon  staff   96B Oct  7 17:56 src
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % head -n 5 data/sample/prices.csv

date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % tail -n 5 data/sample/prices.csv

2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % █
```

TD2 :

Part 1

Which function reads JSON?

import json

Which function reads CSV?

import csv

Which function selects one ticker?

load_instruments()

Which function coordinates execution?

main()

Where are the input files stored?

Path

DATA_DIR

```

(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % src/main.py

zsh: permission denied: src/main.py
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % with open(DATA_DIR / "instruments.json", encoding="utf-8") as file:
{
  instruments = json.load(file)
zsh: unknown file attribute: _
zsh: invalid mode specification
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % print(type(instruments))
zsh: no matches found: print(type(instruments))
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,30000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6590.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
(base) marlon@Marlons-MacBook-Air esilv-marketpulse-g07-TortueNinja % █

```

```
1 import json
2 DATA_DIR = "/Users/marlon/labs/esilv-marketpulse-g07-TortueNinja/data/sample"
✓ [5] 12ms
```

```
1 with open("data/sample/instruments.json", encoding="utf-8") as file:
2     instruments = json.load(file)
✓ [8] 11ms
```

```
1 print(type(instruments))
2 print(instruments.keys())
```

✓ [9] 48ms

```
<class 'dict'>
dict_keys(['instrument', 'benchmark'])
```

```
1 instrument = instruments["instrument"]
2 benchmark = instruments["benchmark"]
3
4 print(instrument["ticker"])
5 print(instrument["name"])
6 print(instrument["currency"])
7
8 print(benchmark["ticker"])
9 print(benchmark["name"])
```

✓ [10] 42ms

```
AAPL
Apple Inc.
USD
SP500
S&P 500
```

```
1 import csv
2 def load_prices():
3     with open("data/sample/prices.csv", encoding="utf-8") as file:
4         return list(csv.DictReader(file))
✓ [20] 45ms
```

```
1 prices = load_prices()
2
3 print(type(prices))
4 print(type(prices[0]))
5 print(prices[0])
✓ [21] 45ms
```

```
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50',
 'low': '248.00', 'close': '250.00', 'volume': '38000000'}
```

```
1 print(prices[0]["close"])
2 print(type(prices[0]["close"]))
✓ [22] 49ms
```

```
250.00
<class 'str'>
```

```
1 close = float(prices[0]["close"])
2
✓ [23] 29ms
```



```
1 def filter_prices(prices, ticker):
2     return [row for row in prices if row["ticker"] == ticker]
✓ [24] 30ms
```

```
1 instrument_prices = filter_prices(  
2     prices,  
3     instrument["ticker"],  
4 )
```

```
5  
6 benchmark_prices = filter_prices(  
7     prices,  
8     benchmark["ticker"],  
9 )
```

✓ [25] 38ms

```
1 print(len(instrument_prices))  
2 print(len(benchmark_prices))
```

✓ [26] 52ms

21

21

```
1 def get_first_close(prices):  
2     first_close_AAPL = prices[0]["close"]  
3     first_close_SP500 = prices[1]["close"]  
4     return float(first_close_AAPL), float(first_close_SP500)  
5  
6 get_first_close(prices)
```

✓ [41] 68ms

(250.0, 6600.0)



```
def get_last_close(prices):  
    last_close_AAPL = prices[-1]["close"]  
    last_close_SP500 = prices[-2]["close"]  
    return float(last_close_AAPL), float(last_close_SP500)
```

```
get_last_close(prices)
```

✓ [44] 43ms

(6742.0, 266.2)

```
def display_market_summary(asset, prices, show_currency=True):  
    rows = [row for row in prices if row["ticker"] == asset["ticker"]]  
  
    print(asset["ticker"], asset["name"])  
    print(len(rows))  
    print(rows[0]["close"])  
    print(rows[-1]["close"])  
  
    if show_currency:  
        print(asset["currency"])
```

✓ [47] 59ms